Introduction to Operating Systems

A computer contains many hardware components such as the processor, memory, disks, keyboard, mouse, display, printers, and network interfaces. Managing all these parts directly is difficult. The **Operating System (OS)** is a special layer of software that makes the computer easier to use and manages all hardware resources efficiently.

What is an Operating System?

An **Operating System** is system software that acts as an interface between the user/application programs and the hardware. It provides a simpler environment for running programs and manages system resources such as CPU, memory, storage, and input/output devices.

Why do we need an Operating System?

A modern computer is a very complex system. It includes:

- One or more processors
- Main memory
- Disks or storage devices
- Keyboard and mouse
- Display/monitor
- Printers
- Network interfaces
- Other input/output devices

If every programmer had to control these devices directly, software development would become extremely difficult. Therefore, the operating system provides:

- a cleaner and simpler model of the computer,
- a way to manage hardware resources,
- a safe environment for running programs,
- efficient coordination among all components.

The OS hides hardware complexity and provides a convenient platform on which application programs can run.

Operating System vs User Interface

Many users think the program they interact with is the operating system. This is not completely correct.

Shell and GUI

The program the user interacts with is usually:

- **Shell** — when the interface is text-based
- **GUI (Graphical User Interface)** — when the interface uses windows, icons, menus, and pointers

Although these programs use the operating system, they are **not the core operating system itself**.

The shell or GUI helps the user interact with the system, but the actual operating system is the lower-level software that controls the hardware and provides services to all programs.

Examples of user programs:

- Web browser
- E-mail reader
- Music player
- Text editor

These programs all depend heavily on the operating system to function properly.

Two Modes of Operation

Most computers operate in two different modes:

1. **Kernel Mode**
2. **User Mode**

Kernel Mode

Kernel Mode (also called **Supervisor Mode**) is the privileged mode in which the operating system runs. In this mode, the OS has complete access to hardware and can execute all machine instructions.

Characteristics of kernel mode:

- Full access to all hardware resources
- Can execute privileged instructions
- Can control memory, CPU, disks, and I/O devices
- Used by the operating system kernel

User Mode

User Mode is the restricted mode in which ordinary application programs run. In this mode, only a subset of machine instructions is allowed.

Characteristics of user mode:

- Limited access to hardware
- Cannot directly perform I/O operations
- Cannot execute privileged instructions
- Must request services from the OS

Comparison

Mode	Kernel Mode	User Mode
Who runs here?	Operating system kernel	Application programs
Hardware access	Full access	Restricted access
Privileged instructions	Allowed	Not allowed
I/O operations	Directly possible	Must go through OS
Purpose	System control and resource management	Running user applications

Why is this separation important?

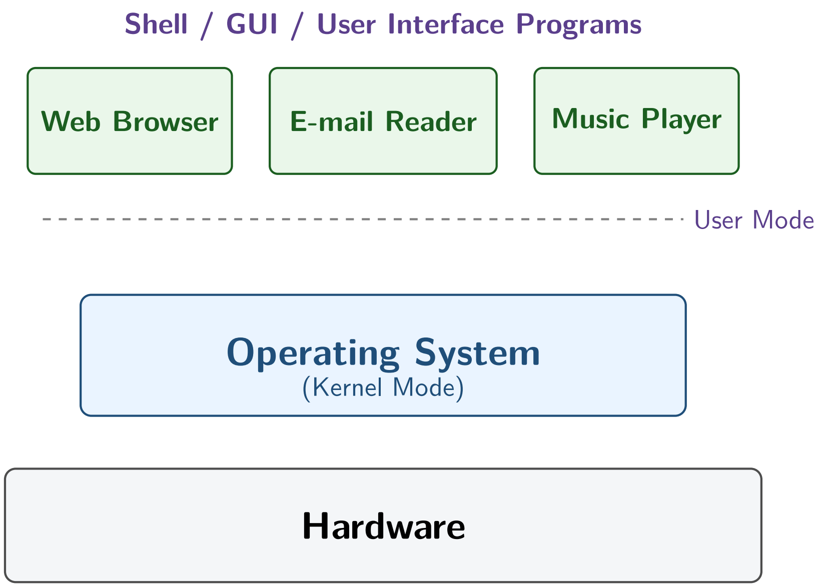
This separation is essential for:

- **Protection** — preventing user programs from damaging the system
- **Security** — preventing unauthorized access to hardware
- **Stability** — preventing one faulty program from crashing everything
- **Control** — ensuring the OS remains in charge of system resources

The OS runs in kernel mode because it must manage and protect the entire system. User applications run in user mode so they cannot directly interfere with critical hardware operations.

Where Does the Operating System Fit?

The operating system sits between hardware and user applications.



Interpretation of the Figure

From bottom to top:

- **Hardware** is the physical part of the computer.
- The **Operating System** runs directly on the hardware.
- **Application Programs** run above the OS in user mode.

- The shell or GUI helps users launch and control these applications.

The operating system provides the foundation on which all other software runs.

OS and Application Programs

There is an important difference between the operating system and normal application software.

Application Programs

A user can:

- install a new browser,
- change the e-mail reader,
- write a new music player,
- replace one application with another.

Operating System Components

A user cannot normally:

- modify the clock interrupt handler,
- rewrite hardware control routines,
- directly change protected kernel functions.

These parts belong to the operating system and are protected by hardware mechanisms.

Application software is usually replaceable by users, but core operating system components are protected and cannot be freely modified.

Boundary of the Operating System

The boundary of the operating system is not always perfectly clear.

Why can the boundary become unclear?

In some systems:

- certain helper programs run in user mode but perform sensitive system tasks,
- parts such as the file system may run in user space,
- embedded systems may not even have a clear kernel/user separation,
- interpreted systems may rely on software rather than hardware for protection.

Example

A password-changing program:

- may run in user mode,
- is not part of the kernel,
- but still performs a sensitive and privileged function.

So, while everything in kernel mode is definitely part of the OS, some user-mode programs may also be closely associated with it.

Characteristics of Operating Systems

Operating systems are very different from ordinary programs.

1. Huge

Modern operating systems are extremely large.

- Core operating systems like Linux or Windows may contain millions of lines of code.
- With libraries included, the total code base can become tens of millions of lines.

2. Complex

An OS must handle:

- process management
- memory management
- file systems
- input/output control

- interrupts
- protection and security
- communication with hardware

3. Long-Lived

Operating systems are not usually discarded and rewritten from scratch. Instead, they evolve gradually over time because:

- they are difficult to design,
- they are costly to replace,
- users need compatibility,
- developers continuously improve them.

Operating systems survive for many years and evolve from version to version rather than being replaced completely.

Examples of Operating System Families

Windows Family

The text mentions that:

- Windows 95 / 98 / Me belonged to one line of development,
- Windows NT / 2000 / XP / Vista / Windows 7 belonged to another.

Even when different versions look similar to users, their internal design may be very different.

UNIX Family

The text also highlights UNIX and its variants:

- System V
- Solaris
- FreeBSD
- Linux

Linux is a separate code base, but it is strongly inspired by UNIX and highly compatible with it.

Operating systems evolve over time. Different versions may appear similar from outside, but their internal architecture can be significantly different.

Conclusion

The operating system is the fundamental software layer of a computer system. It runs directly on hardware, controls and manages system resources, and provides a platform on which all user applications depend. The distinction between kernel mode and user mode is one of the most important concepts in understanding how operating systems work.